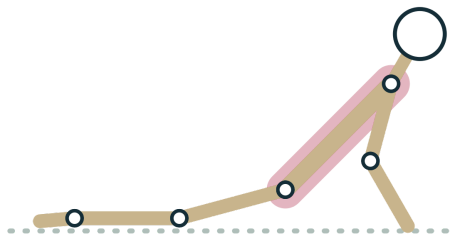


Module 3

Joints: Structure, Range and Stability

Every pose is a set of joint actions, and knowing how joints are built, what limits them and what holds them steady is how you teach real bodies instead of ideal shapes.



2.5 hours • 6 lessons • Dr. Akash Garg, D.C.
Anatomy & Physiology for Vinyasa Teachers

Overview

A joint (also called an articulation) is any place where two or more bones meet. Some joints barely move at all, like the sutures of the skull, while others, like the hip and shoulder, move in every direction. In this module you will learn the three structural classes of joint, the six types of synovial joint and the movements each one allows, and the vocabulary you need to describe any pose as a list of joint actions.

You will also meet two ideas that change how you see a class. First, range of motion ends either by tension (tissue reaching its stretch limit) or by compression (something running into something else), and bone shape differs from person to person, so two students may never reach the same shape. Second, joints are held stable passively by bone fit, capsule and ligaments, and actively by muscle. That is why very flexible students need strength and control, and why stiff students need patience, props and respect rather than force. Teachers adapt and refer; they never diagnose.

By the end of this module you can

1. Classify a joint as fibrous, cartilaginous or synovial and give a body example of each.
2. Name the six synovial joint types, give an example of each, and state the movements each allows.
3. Describe any yoga pose as a list of joint actions using flexion, extension, abduction, adduction, rotation and related terms.
4. Distinguish tension from compression as the end of a range of motion and explain how individual bone shape limits a pose.
5. Explain passive and active joint stability and compare the hip with the shoulder.
6. Recognise hypermobile and hypomobile patterns in class and choose safe cues and props without diagnosing.

Lessons

- **3.1** Fibrous, cartilaginous and synovial joints (20 min)
- **3.2** The six synovial joint types and the movements each allows (30 min)
- **3.3** Range of motion: tension versus compression and individual bone shape (30 min)
- **3.4** Joint stabilization: passive (ligament, capsule, bony fit) and active (muscle) (30 min)
- **3.5** Hypermobility and hypomobility in the yoga room (20 min)
- **3.6** Pose lab: reading every joint action in Sun Salutation A (20 min)

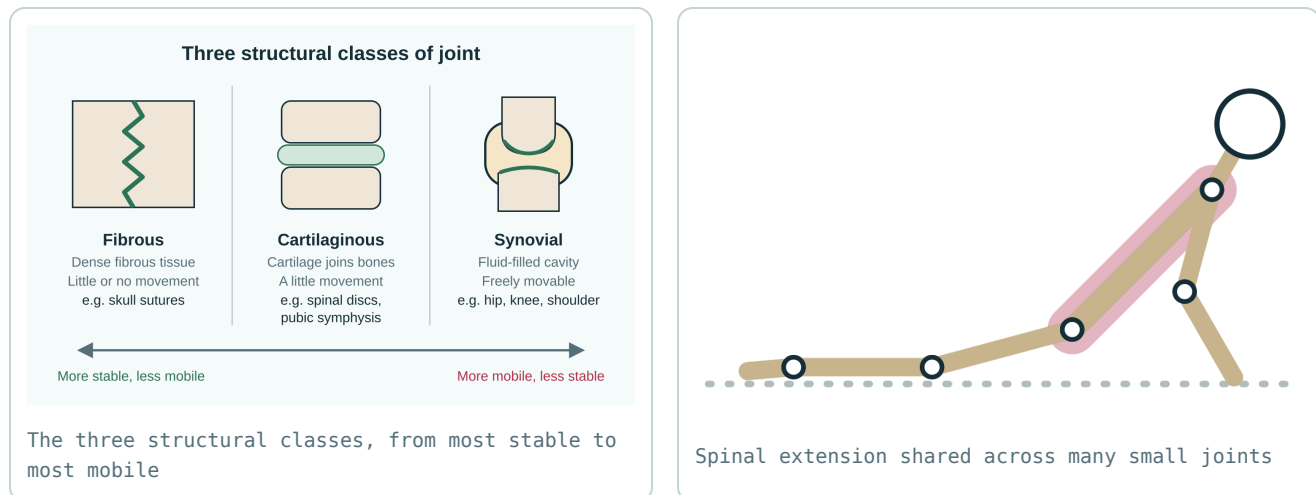
3.1 Fibrous, cartilaginous and synovial joints

A joint, or articulation, is any place where two or more bones meet. Anatomists sort joints in two ways. The structural classification asks what tissue connects the bones. The functional classification asks how much the joint moves: immovable (synarthrosis), slightly movable (amphiarthrosis) or freely movable (diarthrosis). The two systems line up closely, which makes them easy to learn together.

Fibrous joints are joined by dense fibrous connective tissue and have no joint cavity. Examples are the sutures of the skull, the tooth in its socket (a gomphosis) and the lower tibia and fibula, bound by a ligament (a syndesmosis). They allow little or no movement. Cartilaginous joints are joined by cartilage and also lack a cavity. The intervertebral discs and the pubic symphysis are made of fibrocartilage and allow a small amount of movement. The growth plates of young bones are temporary cartilaginous joints made of hyaline cartilage. Synovial joints have a fluid-filled cavity enclosed by a capsule, and they move freely. The hip, knee, ankle, shoulder, elbow and wrist are all synovial, which is why most of this module focuses on them.

The spine is a good example of how small movements add up. The adult spine has 23 intervertebral discs. Each disc, together with the small synovial facet joints behind it, allows only a few degrees of movement, yet shared across the whole column they produce the large curves of Cat, Cow, Cobra and twists. The sacroiliac joint is a mixed case: synovial at the front, bound by very strong ligaments behind, and moving only a few degrees. Its main job is load transfer between spine and legs, so be cautious with popular claims that it goes out of place, and refer students with pain in this area.

The key pattern to remember is a trade-off. As joints become more mobile, they rely less on their own structure for stability and more on the muscles around them.



KEY POINTS

- A joint (articulation) is any place where bones meet.
- Fibrous joints: dense fibrous tissue, little or no movement, e.g. skull sutures.
- Cartilaginous joints: cartilage link, slight movement, e.g. intervertebral discs and pubic symphysis.
- Synovial joints: fluid-filled capsule, free movement, e.g. hip, knee, shoulder.
- The spine's range comes from many small joints each contributing a little.
- More mobility usually means less built-in stability.

FEEL IT

Stand in Mountain. Press gently on your skull and notice there is no give at the sutures. Slowly round and arch your spine and feel many small joints share the movement. Then lift one knee and circle the thigh, feeling how freely the synovial hip moves. Write down one pose that mainly uses the spine and one that mainly uses the hip.

CUE IT

“Let the whole spine share the backbend, a little from every level rather than a lot from one place.”

“Move from the hip joint itself, the crease at the top of the thigh, as you fold forward.”

“If the front of the pelvis feels tender, bring your feet a little closer together.”

CHECK IT

Which joint is cartilaginous?

Answer: The pubic symphysis. The pubic symphysis joins the two pubic bones with a fibrocartilage disc and allows only slight movement.

3.2 The six synovial joint types and the movements each allows

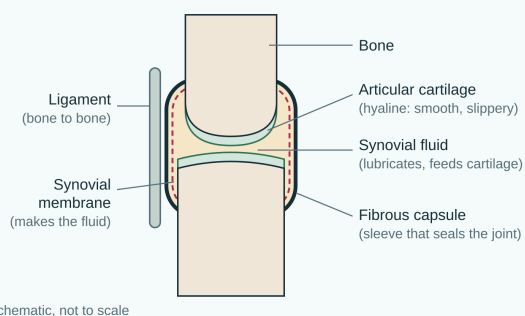
Synovial joints share a common design. The bone ends are capped with smooth articular (hyaline) cartilage. A fibrous capsule encloses the joint, and its inner synovial membrane secretes synovial fluid, which lubricates the joint and nourishes the cartilage, since cartilage has no blood vessels of its own. Ligaments join bone to bone and limit extreme movement. Some joints add structures such as the knee's menisci or the labrum that deepens the hip and shoulder sockets.

Synovial joints are named by the shape of their surfaces, and shape determines movement. Hinge joints (elbow, ankle, finger joints) move around one axis: flexion and extension. The knee is a modified hinge that can also rotate a little when bent. Pivot joints (the first two neck vertebrae, the radioulnar joints) rotate around one axis. Condylloid joints (wrist, knuckles) move in two axes but cannot truly rotate. The saddle joint at the base of the thumb moves in two axes and allows opposition. Plane joints (between the small wrist and foot bones, the spinal facets) allow small glides. Ball-and-socket joints (hip and shoulder) move around three axes, including rotation.

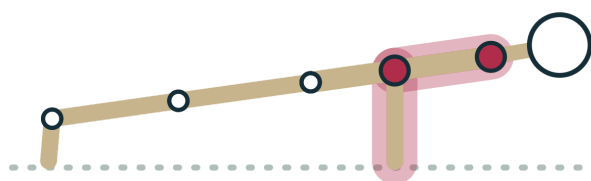
To read a pose you need the movement vocabulary. In the sagittal plane: flexion (the joint angle closes) and extension (it opens), plus dorsiflexion and plantarflexion at the ankle. In the frontal plane: abduction (away from the midline), adduction (toward it) and lateral flexion of the spine. In the transverse plane: internal and external rotation, spinal rotation, and pronation and supination of the forearm. Circumduction combines these into a circle.

In practice, most poses combine several actions. The front hip in Warrior II is flexed, abducted and externally rotated all at once. Because the pelvis links the two hips, changing pelvis position changes the demand on each hip, which is why flexible alignment choices serve students better than one fixed rule.

Anatomy of a synovial joint



Parts shared by every synovial joint



Chaturanga: elbow flexion held by strong muscles

KEY POINTS

- Synovial joints share cartilage, capsule, synovial membrane, fluid and ligaments.
- Hinge and pivot: one axis. Condylloid and saddle: two axes. Ball-and-socket: three axes.
- Plane joints glide a little; the knee is a modified hinge.
- Flexion closes a joint angle, extension opens it; abduction moves away from the midline.
- Rotation happens in the transverse plane; the forearm pronates and supinates.
- Most poses combine several joint actions, and the pelvis links both hips.

FEEL IT

Sit or stand and take each joint type through its menu: bend and straighten the elbow, turn the palm up and down, move the wrist in two directions, touch the thumb to each fingertip, then lift one leg and move the hip in three planes. For each, say the joint type and the movement names aloud. Finish by naming the three actions of the front hip in Warrior II.

CUE IT

“Let the front knee point in the same direction as your middle toes.”

“Turn your palms up and feel the forearm bones rotate, not the wrist.”

“Find the angle for your back foot that lets your front hip feel easy.”

CHECK IT

A student turns their palms up in Savasana. Which joint type makes this possible?

Answer: Pivot joints between the radius and ulna. Turning the palm up (supination) happens as the radius pivots around the ulna at the radioulnar joints.

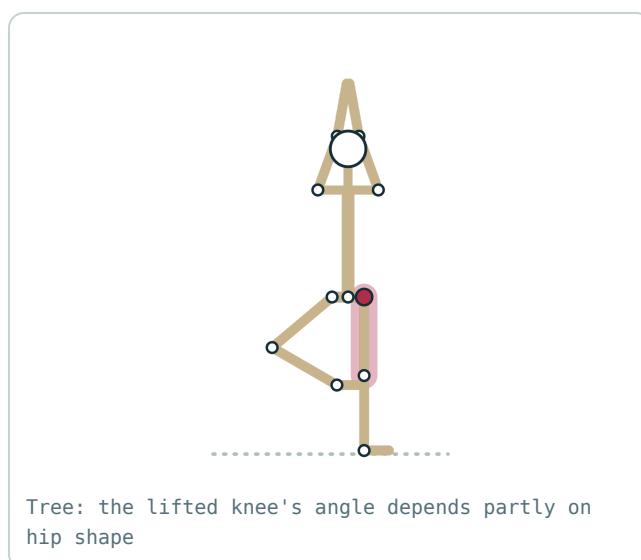
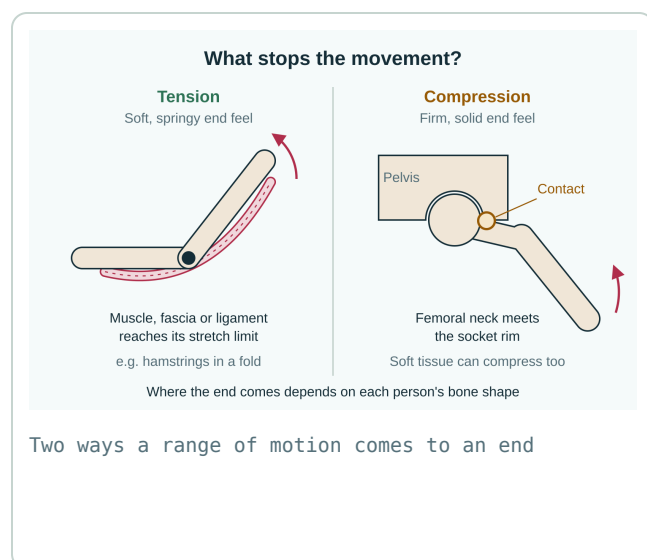
3.3 Range of motion: tension versus compression and individual bone shape

Range of motion (ROM) is how far a joint can move in a given direction. Every range has an end, and the most useful question a teacher can ask is: what is creating that end? A simple framework, popularised in yoga by teachers such as Paul Grilley and Bernie Clark, sorts ends into two kinds. Tension means a tissue on the opposite side of the movement has reached its stretch limit: muscle, fascia, capsule or ligament. It feels soft and springy and is sensed as a stretch away from the moving joint. Compression means something has met something else: bone on bone, or soft tissue on soft tissue. It feels firm and is sensed where the parts meet, often as a pinch.

The difference matters because the two respond differently to practice. Regular stretching often increases range limited by tension, partly through tissue adaptation and substantially through increased tolerance of the stretch sensation. Bony compression does not change with stretching, and pushing into it may irritate the joint. Soft tissue compression depends on body size and shape.

Individual bone shape is the key reason two students cannot always reach the same shape. The femoral neck meets the shaft at an average angle of about 125 degrees in adults, with wide variation either side. The neck also twists forward by differing amounts (femoral version), and hip sockets vary in depth and orientation. Together these decide where compression ends hip rotation and abduction. Similar variation exists at the shoulder, elbow and spine. No amount of effort reshapes adult bone.

For teaching, this shifts the goal from matching a photo to serving the purpose of the pose. Cue sensation: stay and breathe with a stretch; change the angle or add a prop with a pinch or hard stop. Offer options, and never press a student past a compression end.



KEY POINTS

- Range of motion ends by tension (stretch limit) or compression (parts meeting).
- Tension is felt opposite the movement; compression is felt where parts meet.
- Tension-limited range often improves with practice; bony compression does not.
- Bone shape, such as femoral neck angle and version, varies widely between people.
- Judge a pose by its purpose, not by how closely it matches a photo.
- A pinch or hard stop calls for a new angle or a prop, not more effort.

FEEL IT

Lie on your back and hug one knee to your chest, noting whether the limit is a stretch behind or a pinch in front. Move the knee toward the same-side armpit and notice whether the pinch eases. Then compare Child's Pose with knees together and knees wide. Label each end as tension or compression and note which change helped.

CUE IT

"If you feel a stretch, stay and breathe. If you feel a pinch, change the angle."

"Let your knee point where your hip allows; the balance is the pose."

"Bend your knees as much as you need to fold from the hips without a pinch."

CHECK IT

In a deep squat, a student feels a hard pinch at the front of the hip. What is the most likely limit?

Answer: Compression at the front of the hip. A pinch felt where parts meet, at the front of the hip in deep flexion, points to compression, which stretching will not change.

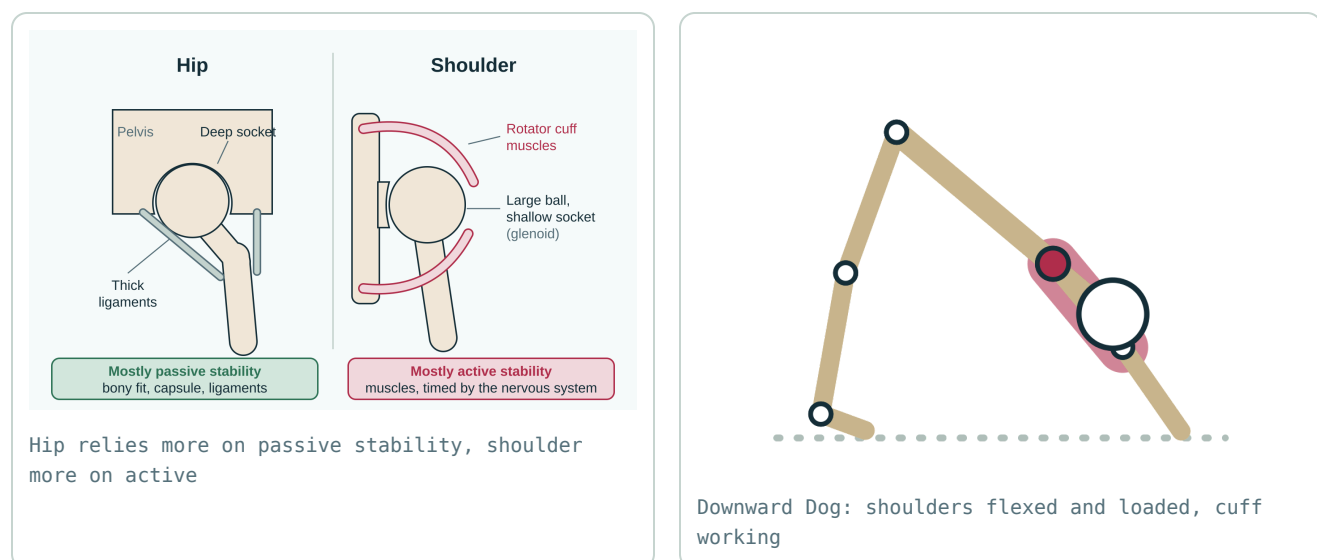
3.4 Joint stabilization: passive (ligament, capsule, bony fit) and active (muscle)

Joint stability is the ability to keep joint surfaces centred and resist unwanted movement under load. It comes from two sources. Passive stabilizers work without effort: the bony fit of the joint surfaces, the fibrous capsule, the ligaments, and structures such as the labrum (a fibrocartilage rim deepening the hip and shoulder sockets) and the menisci (C-shaped fibrocartilage pads in the knee). Active stabilizers are the muscles that cross a joint, switched on and timed by the nervous system using information from sensors in muscles, tendons and joints.

Every joint uses both, in different proportions. The hip bears body weight, so it has a deep socket, a labrum, and very strong ligaments: it relies heavily on passive stability. The shoulder is built for reach. Its large humeral head sits on a small, shallow glenoid, so it has the greatest range of any joint but little bony fit. It relies on the four rotator cuff muscles (supraspinatus, infraspinatus, teres minor and subscapularis) to keep the head centred. Weight-bearing on the hands, as in Plank, Downward Dog and arm balances, asks a reaching joint to act like a supporting one, so it needs strength and gradual progression.

The knee sits between these extremes. The femur rests on a relatively flat tibia, so the bony fit is poor. The cruciate and collateral ligaments guide it, the menisci deepen and cushion it, and the quadriceps, hamstrings and hip muscles add active control. The main risk in yoga is a bent, loaded knee twisting while the foot stays fixed.

Passive structures cannot be switched on, and overstretched ligaments do not readily return to their original length. Active stability, by contrast, is trainable through strength, balance and control. This is the anatomy behind cues such as soften the knees, press the floor away and engage the legs: they move load from passive end range onto muscle.



KEY POINTS

- Passive stability: bony fit, capsule, ligaments, labrum, menisci.
- Active stability: muscles, coordinated by the nervous system.
- The hip is mostly passively stable; the shoulder is mostly actively stable.
- The rotator cuff keeps the humeral head centred on the shallow glenoid.
- The knee combines ligaments, menisci and muscle; avoid loaded twisting of a bent knee.
- Active stability is trainable; overstretched ligaments do not readily recover.

FEEL IT

Stand in Tree and watch the constant small corrections of your standing foot and ankle. Then come into Plank, first letting the chest sink between the shoulder blades, then pressing the floor away. Notice how the shoulders feel in each version. Finish in Mountain, locking the knees back and then softening them with legs engaged, and compare.

CUE IT

“Press the floor away and let the muscles around your shoulders switch on.”

“Soften your knees a touch and firm the muscles of your legs.”

“Keep your front knee tracking toward your middle toes as you bend it.”

CHECK IT

Why does the shoulder depend more on muscle for stability than the hip does?

Answer: Its large head sits on a small, shallow socket. The shallow glenoid gives little bony fit, so the rotator cuff must actively keep the humeral head centred.

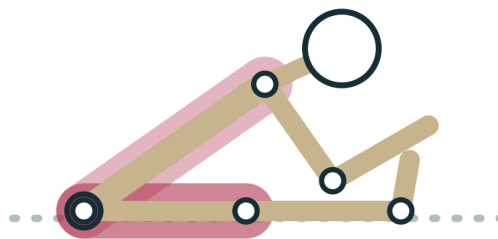
3.5 Hypermobility and hypomobility in the yoga room

Joint range sits on a spectrum. Hypermobile joints move beyond the typical range; hypomobile joints move less. Many people have a mix, and neither end is good or bad in itself. Generalised joint hypermobility is common, more so in younger people and in women, and often causes no symptoms. Yoga classes tend to attract flexible people, so you will meet many. Hypomobility can come from bone shape, tissue stiffness, age, past injury or conditions such as arthritis.

The Beighton score is a nine-point screening tool that clinicians use to assess generalised joint hypermobility. It awards one point per side for bending the little finger back beyond 90 degrees, bringing the thumb to the forearm, elbow hyperextension beyond 10 degrees and knee hyperextension beyond 10 degrees, and one point for placing the palms flat on the floor with straight knees. Thresholds depend on age. A high score indicates hypermobility, not a disorder. Clinicians also recognise hypermobility spectrum disorders and hypermobile Ehlers-Danlos syndrome, which involve features such as pain, fatigue or joint instability and are diagnosed through full clinical assessment. Yoga teachers do not score, screen or diagnose students.

In class, very mobile students may lock the knees or elbows past straight and rest in end range on their ligaments and capsule. They benefit from micro-bends, working short of end range, strength and balance work, and a slower pace. Stiff students reach end range early, may strain or hold the breath, and benefit from props, bent knees, time and a clear message that their version is complete.

The common thread is adapting the practice rather than labelling the person. Cue control over depth for the whole room. If a student reports ongoing pain, fatigue, frequent joint slips or dislocations, suggest they see a doctor or physiotherapist. Teachers adapt and refer; they never diagnose.



The same fold can be effortless for one student, out of reach for another

KEY POINTS

- Hypermobility and hypomobility are ends of a spectrum; many people have a mix.
- The Beighton score is a 9-point clinical screening tool, not a diagnosis.
- Hypermobility spectrum disorders and hEDS are diagnosed by clinicians, not teachers.
- Very mobile students need control, strength and work short of end range.
- Stiff students need props, time and freedom from comparison.
- Refer students with pain, fatigue or frequent joint instability.

FEEL IT

Come into Warrior I and bend the front knee as far as is comfortable, then rise a few centimetres until the legs feel strongly engaged. Hold for five breaths and compare. Repeat the idea in Downward Dog with the elbows softened slightly. Note which version felt steadier and which you would cue for a very flexible student.

CUE IT

“Find your deepest point, then come back a little and hold it with your muscles.”

“Keep a whisper of bend in your elbows and press through the base of your index fingers.”

“Use a block to bring the floor closer; your version of the pose is complete.”

CHECK IT

A student says they are very flexible and asks you to test their Beighton score. What should you do?

Answer: Explain it is a clinical tool, offer control-focused options, suggest a health professional if they have symptoms. Teachers do not screen or diagnose; they adapt the practice and refer when there are symptoms.

3.6 Pose lab: reading every joint action in Sun Salutation A

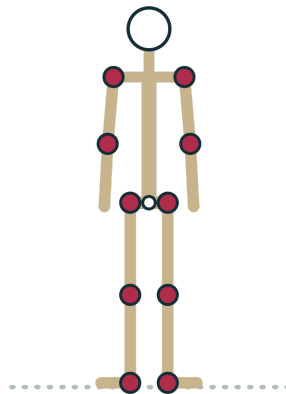
Reading a pose means describing it as a set of joint actions. A reliable method is to take one joint at a time, name the plane of movement, name the action compared with Mountain (the neutral reference), then ask what limits the range (tension or compression) and what stabilizes the joint (passive structures, muscle or both). Practise this on Sun Salutation A until it becomes quick.

From Mountain, Upward Salute brings the shoulders into flexion, with upward rotation of the shoulder blades and extended elbows. Forward Fold is mainly hip flexion with extended knees, plus spinal flexion as the fold deepens; tension in the hamstrings is the usual limit, so bent knees are a sensible option. The half lift keeps the hips flexed while the spine lengthens back toward neutral.

In Plank the shoulders are flexed about 90 degrees relative to the trunk, the elbows extended and the ankles dorsiflexed with the toes extended on the floor. Lowering to Chaturanga flexes the elbows toward 90 degrees and moves the shoulders toward extension. The shoulder is weight-bearing and relies on active stability, while the abdominal and hip muscles hold the hips and spine near neutral. Knees down reduces the load.

Cobra and Upward-facing Dog are mainly spinal extension, shared across many small joints, with hip extension and ankle plantarflexion; the elbows stay bent in Cobra and extend in Up Dog. A pinch in the low back suggests compression and calls for less height. Downward-facing Dog combines full shoulder flexion under load, hip flexion, knee extension and ankle dorsiflexion. Hamstring and calf tension are common limits, and heels need not reach the floor. The sequence then reverses through half lift, fold and Upward Salute back to Mountain.

Sun Salutation A takes nearly every major joint through a range, which makes it a good warm-up and an ideal training ground for your eye.



Mountain: the neutral reference for every joint

KEY POINTS

- Read a pose joint by joint: plane, action, limit, stabilizer.
- Upward Salute: shoulder flexion, elbow extension.
- Forward Fold: hip flexion, knee extension, spinal flexion; hamstring tension limits it.
- Chaturanga: elbow flexion and shoulder extension under load; knees down reduces it.
- Cobra and Up Dog: spinal and hip extension, ankle plantarflexion.
- Downward Dog: shoulder and hip flexion, knee extension, ankle dorsiflexion.

FEEL IT

With a partner, move through one round of Sun Salutation A at half speed. At each pose, pause and name three joint actions aloud while your partner checks the plane, action and likely limit. Swap roles, then write down any action you disagreed about and look it up in this module.

CUE IT

“Bend your knees as much as you need so the fold comes from your hips.”

“Lower only as far as your elbows and shoulders stay steady, knees down is a strong choice.”

“Share the backbend along your whole spine; if your low back pinches, come a little lower.”

CHECK IT

Moving from Plank into Upward-facing Dog, what happens at the ankles?

Answer: They move from dorsiflexion to plantarflexion. In Plank the toes are tucked and the ankles dorsiflexed; in Up Dog the tops of the feet press down, so the ankles plantarflex.

Key terms

Joint (articulation)	Any place where two or more bones meet, such as the hip where the femur meets the pelvis in Warrior II.
Fibrous joint	Bones joined by dense fibrous tissue with little or no movement, like the skull sutures that stay still in every headstand prep.
Cartilaginous joint	Bones joined by cartilage that allows slight movement, like the intervertebral discs that each add a little to Cat and Cow.
Pubic symphysis	The fibrocartilage joint at the front of the pelvis; it can feel tender in wide-legged poses, especially in pregnancy.
Synovial joint	A freely movable joint with a fluid-filled cavity inside a capsule, such as the shoulder in Downward Dog.
Articular cartilage	The smooth hyaline cartilage covering bone ends inside a synovial joint, letting the knee glide as you bend into Chair.
Synovial fluid	Thick fluid inside a synovial joint that lubricates it and nourishes the cartilage; movement such as a gentle warm-up helps circulate it.
Joint capsule	The fibrous sleeve enclosing a synovial joint; locking the knees back in Mountain rests the joint on the back of its capsule.
Ligament	A band of dense connective tissue joining bone to bone, limiting extreme range, like the strong hip ligaments that tighten as you stand upright.
Hinge joint	A synovial joint moving in one axis, flexion and extension, like the elbow bending in Chaturanga.
Pivot joint	A synovial joint that rotates around one axis, like the neck turning to look up in Triangle.
Ball-and-socket joint	A synovial joint moving in three axes including rotation, like the hip rotating outward in Tree.
Flexion and extension	Flexion closes a joint angle and extension opens it; Forward Fold flexes the hips, Cobra extends the spine.
Abduction and adduction	Abduction moves a limb away from the midline and adduction toward it; the front hip abducts in Warrior II.
Dorsiflexion and plantarflexion	Dorsiflexion brings the top of the foot toward the shin, as in Downward Dog; plantarflexion points the foot, as in Cobra.
Range of motion	How far a joint can move in a given direction, such as how far the thigh lifts toward the chest in a knee hug.
Tension	An end of range created by tissue reaching its stretch limit, felt as a stretch opposite the movement, like the hamstrings in a fold.

Compression	An end of range created by parts meeting, bone on bone or soft tissue on soft tissue, felt as a pinch, like the front of the hip in a deep squat.
Passive stability	Support from structures that need no effort: bony fit, capsule, ligaments, labrum and menisci, like the deep hip socket in standing poses.
Active stability	Support from muscles switched on by the nervous system, like the rotator cuff holding the shoulder steady in Plank.
Hypermobility	Joint range beyond the typical; common and often harmless, but very mobile students benefit from micro-bends and control, for example in Downward Dog.
Beighton score	A 9-point clinical screening tool for generalised joint hypermobility; it is not a diagnosis, and teachers do not score students.

Myths to retire

With enough practice, anyone can reach the full expression of any pose.

Bone shape varies between people and ends some ranges by compression. Stretching cannot reshape adult bone, so some shapes are simply not available to some bodies.

More flexibility always means healthier joints.

Very mobile joints can rely heavily on ligaments and capsule at end range. They benefit from strength and control, and flexible students are not protected from injury.

A pinch in a pose means you need to stretch harder.

A pinch usually signals compression, where parts meet. More effort will not change it; a different angle or a prop is the right response.

The sacroiliac joint regularly slips out of place and a teacher can put it back.

The SI joint moves only a few degrees and is held by very strong ligaments. Pain here needs a qualified health professional, not a manual adjustment from a yoga teacher.

Locking the knees straight is the most stable way to stand.

Locking rests the knee on its capsule and ligaments. A slight softening with active leg muscles shares the load with muscle.

Self check

1. What is the difference between the structural and functional classifications of joints?
2. Name the six synovial joint types and give one example of each.
3. Describe the joint actions of the front hip in Warrior II.
4. How can a student tell whether their range is limited by tension or compression?
5. Why might two dedicated students never reach the same shape in Lotus or Tree?
6. Compare the stability of the hip and the shoulder.
7. What is the Beighton score and what is the teacher's role regarding it?
8. List the joint actions at the shoulders, hips, knees and ankles in Downward-facing Dog.

Model answers

1. Structural classification describes the tissue joining the bones: fibrous, cartilaginous or synovial. Functional classification describes how much the joint moves: immovable, slightly movable or freely movable.
2. Hinge (elbow), pivot (C1 on C2), condyloid (wrist), saddle (base of thumb), plane (between carpal bones) and ball-and-socket (hip).
3. The front hip is flexed, abducted and externally rotated at the same time. How much rotation it needs depends partly on how the pelvis is positioned.
4. Tension is felt as a stretch on the side opposite the movement, with a springy quality. Compression is felt where parts meet, often as a pinch or hard block.
5. Hip bone shape, including femoral neck angle, femoral version and socket depth and orientation, varies between people. These differences change where compression ends the range, and stretching cannot change adult bone.
6. The hip has a deep socket, labrum and strong ligaments, so it relies mostly on passive stability. The shoulder's large head on a shallow glenoid relies mostly on active stability from the rotator cuff.
7. It is a 9-point clinical screening tool for generalised joint hypermobility. Teachers do not score or diagnose students; they adapt the practice toward control and strength and refer students with symptoms.
8. Shoulders flexed, hips flexed, knees extended as far as the hamstrings allow, and ankles dorsiflexed.